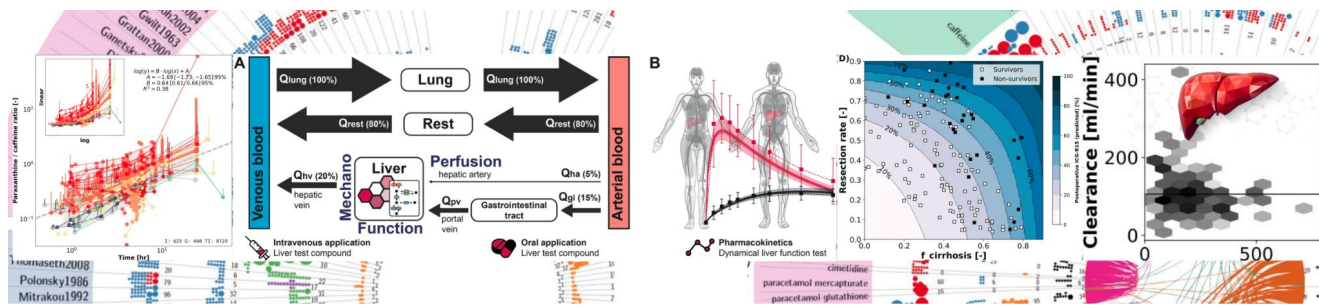




Pharmacokinetics Modeling Course: SBML

Matthias König, MB19, SS2024

Humboldt-University Berlin, Systems Medicine of the Liver Lab

<https://livermetabolism.com>

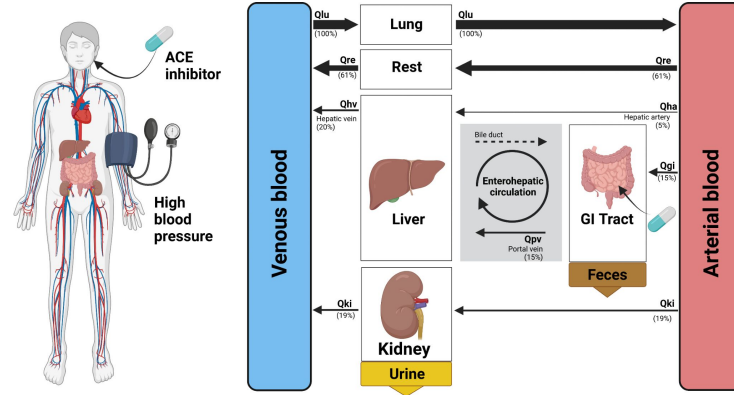
 konigmatt



Systems Biology Markup Language

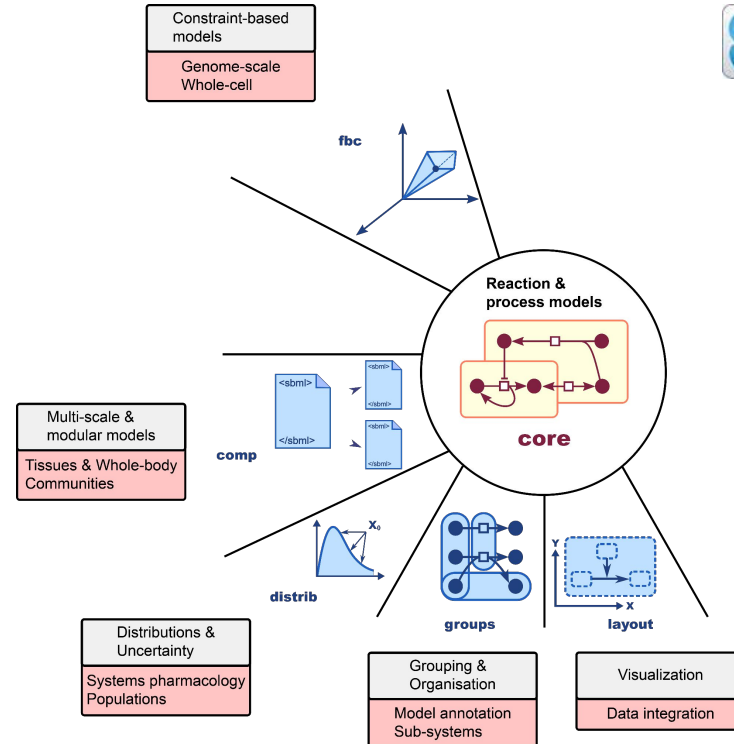


- SBML is a software data format for describing models in biology (<https://sbml.org>)
- It's a little like HTML but for model instead of web pages
- independent of any particular tool
- free and open
- De facto standard

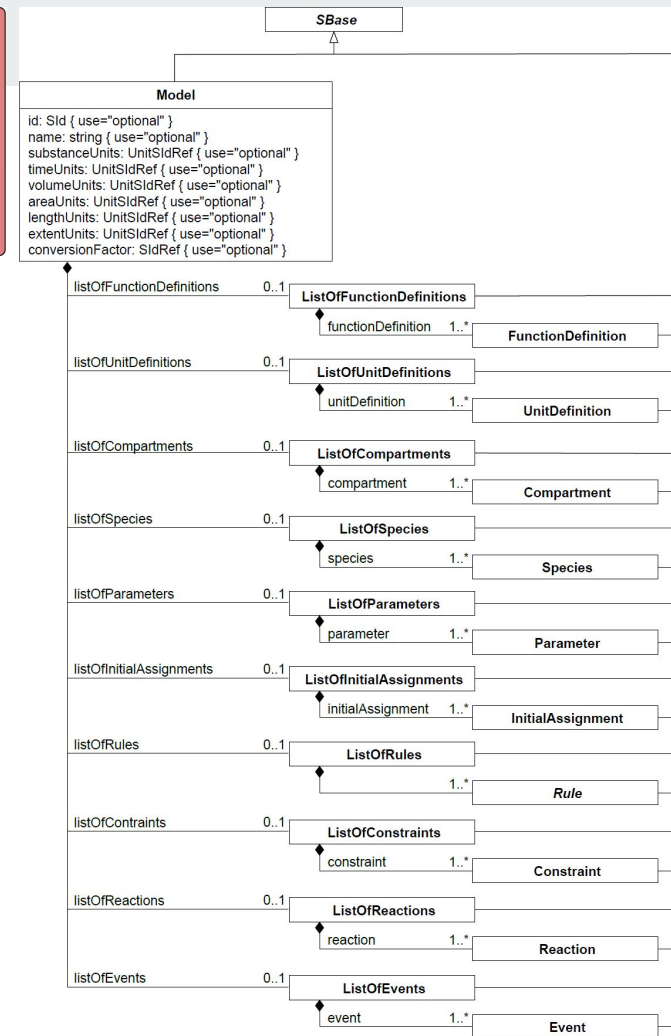
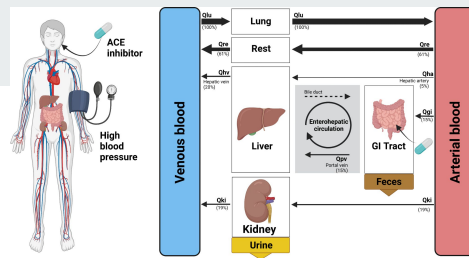


SBML

- **Process-based ODE models**
- Reproducible & exchangeable model encoding (**SBML**)
- Annotations to modelling, biological and medical ontologies (**SBML core**)
- Hierarchical models/multi-scale models (**SBML comp**)
- Unit validation, unit checking, unit conversion
- Distributions in models & uncertainty in data and parameters (**SBML distrib**)
- Mass- & charge balance (**SBML fbc**)
- Use wide range of tools (visualization, parameter fitting, simulation, ...)
- <http://sbml.org>



Keating SM, Waltemath D, König M, ... , Hucka M; SBML Level 3 Community members. SBML Level 3: an extensible format for the exchange and reuse of biological models. Mol Syst Biol. 2020 Aug;16(8):e9110. doi: 10.15252/msb.20199110.



SBML core

- Compartments
 - containers/volumes (e.g. liver volume)
- Species
 - molecules in compartments (e.g. glucose in plasma)
- Parameters
 - parameters with values (e.g. cardiac output)
- AssignmentRules
 - mathematical relationships between other parameters, species, compartments (e.g. $BMI = \text{bodyweight} / \text{height}^2$)
- Reactions
 - processes converting species in other species (e.g. enzymatic conversion; e.g. transport via blood flow)

sbmlutils

Python utilities for **working with SBML models**
<https://github.com/matthiaskoenig/sbmlutils>

Packages

- core, fbc, comp, distrib, layout

Features

- model creation, manipulation & merging
- unit support
- model annotation
- interpolations
- file converters (XPP)

New

- Simplified unit support (unit code completion)
- Markdown annotations
- OMEX support
- Resolving/rendering/validating OMEX metadata
- Interactive model reports (<https://sbml4humans.de>)

```
Parameter(  
    "ICGIM_ki_bil",  
    0.02,  
    unit=U.mM,  
    name="Ki bilirubin of icg import",  
    sboTerm=SBO.INHIBITORY_CONSTANT,  
    notes=""  
    bilirubin reference range:  
    ~ 0.1 - 5 [g/dl]  
    (10 [mg/l] / 584.6623 [g/mole]) ~ 0.0171 mmole/l  
    setting Ki in reference range  
    """,  
    ),
```

Parameter

ICGIM_ki_bil Ki bilirubin of icg import

id	ICGIM_ki_bil
metaID	meta_ICGIM_ki_bil
name	Ki bilirubin of icg import
sbo	SBO:0000261
value	0.02
constant	✓
units	$\frac{\text{mmol}}{\text{l}}$
derivedUnits	$\frac{\text{mmol}}{\text{l}}$
cvterms	
	BQB_IS sbo SBO:0000261
	inhibitory constant
	Synonym: Ki

notes

bilirubin reference range:

~ 0.1 - 5 [g/dl]
(10 [mg/l] / 584.6623 [g/mole]) ~ 0.0171 mm

setting Ki in reference range



SBML4Humans

<https://sbml4humans.de>

- **interactive SBML report** with navigation between SBML objects
- **web application** (no setup)
- **search** and filter functionality
- **resolve/render metadata**
- **hierarchical models** (SBML comp)
- **distributions and uncertainties** (SBML distrib)
- **flux balance** (SBML fbc)
- **COMBINE archives** (multiple models)
- **URL endpoint** for integration in tools/workflows/ webpages/ presentations

https://sbml4humans.de/model_url?url=https://www.ebi.ac.uk/biomodels/model/download/BIOMD0000000001.2?filename=BIOMD0000000001_url.xml

Parameter

HCT hematocrit

id HCT
metaID meta_HCT
name hematocrit
sbo SBO:0000002
value 0.51
constant
units —
derivedUnits —
cvterms

BQB_IS sbo SBO:0000002

quantitative systems description parameter

A numerical value that defines certain characteristics of systems or system functions. It may be part of a calculation, but its value is not determined by the form of the equation itself, and may be arbitrarily assigned.

BQB_IS ncit C64796

Hematocrit Measurement

A measure of the volume of red blood cells expressed as a percentage of the total blood volume. Normal in males is 43-49%, in females 37-43%.

Synonyms

- HCT
- Packed Cell Volume
- Hematocrit
- Erythrocyte Volume Fraction
- PCV
- Hematocrit Measurement
- EVF

BQB_IS omit 0007571

Hematocrit

BQB_IS efo 0004348

hematocrit

The volume of packed RED BLOOD CELLS in a blood specimen. The volume is measured by centrifugation in a tube with graduated markings, or with automated blood cell counters. It is an indicator of erythrocyte status in disease.

Parameter (1)

id	name	constant	value	units	derivedUnits	assignment
HCT	hematocrit		0.51	—	—	

AssignmentRule (8)

id	name	variable	math	derivedUnits
Vve	Vve	Vve	$(1 - HCT) \cdot (BW \cdot FVve - \frac{FVve}{FVar + FVve + FVpo + FVhv} \cdot BW \cdot Fblood \cdot (1 - (FVar + FVve + FVpo + FVhv)))$	<i>t</i>
Var	Var	Var	$(1 - HCT) \cdot (BW \cdot FVar - \frac{FVar}{FVar + FVve + FVpo + FVhv} \cdot BW \cdot Fblood \cdot (1 - (FVar + FVve + FVpo + FVhv)))$	<i>t</i>
Vpo	Vpo	Vpo	$(1 - HCT) \cdot (BW \cdot FVpo - \frac{FVpo}{FVar + FVve + FVpo + FVhv} \cdot BW \cdot Fblood \cdot (1 - (FVar + FVve + FVpo + FVhv)))$	<i>t</i>
Vhv	Vhv	Vhv	$(1 - HCT) \cdot (BW \cdot FVhv - \frac{FVhv}{FVar + FVve + FVpo + FVhv} \cdot BW \cdot Fblood \cdot (1 - (FVar + FVve + FVpo + FVhv)))$	<i>t</i>
Vre_plasma	Vre_plasma	Vre_plasma	$Vre \cdot Fblood \cdot (1 - HCT)$	<i>t</i>
Vgi_plasma	Vgi_plasma	Vgi_plasma	$Vgi \cdot Fblood \cdot (1 - HCT)$	<i>t</i>
Vli_plasma	Vli_plasma	Vli_plasma	$Vli \cdot Fblood \cdot (1 - HCT)$	<i>t</i>
Vlu_plasma	Vlu_plasma	Vlu_plasma	$Vlu \cdot Fblood \cdot (1 - HCT)$	<i>t</i>

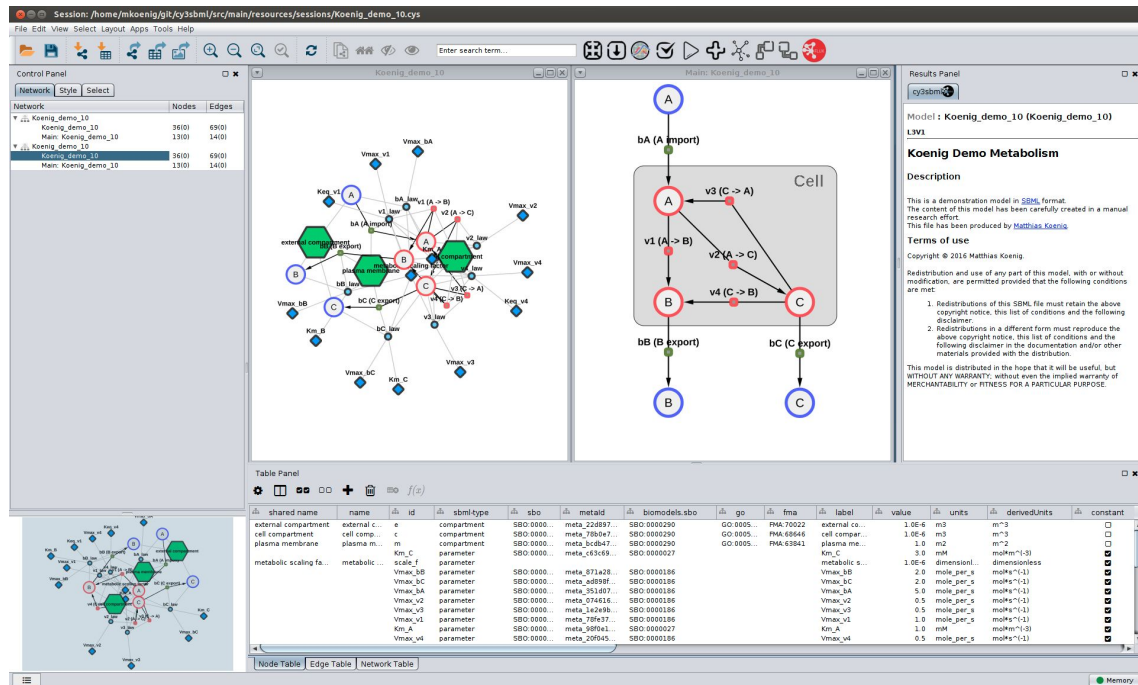
cy3sbml

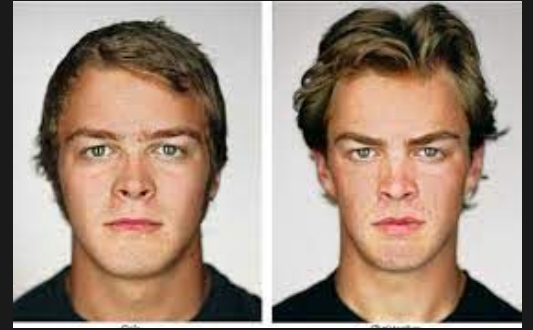
Cytoscape app for visualizing SBML models

<https://github.com/matthiaskoenig/cy3sbml>

Features

- kinetic & reaction-species view
- subgraphs & filtering
- annotation support
- works for large scale networks (genome-scale)
- sbmlutils integration (py2cytoscape)

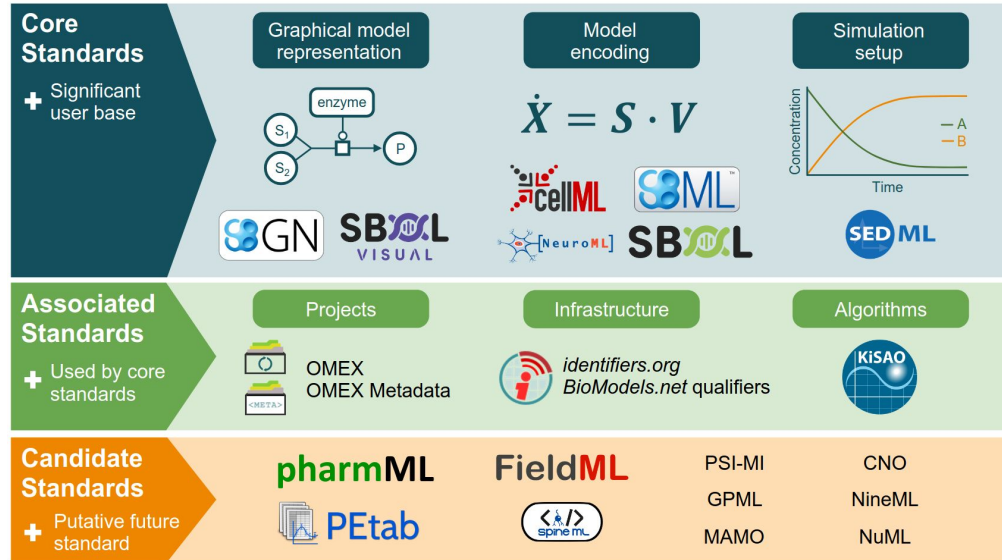




REPRODUCIBILITY



Development of standards



- Models (SBML)
- Simulationen (SED-ML)
- Visualisation (SGBN)
- Parameter Optimization (PETab)
- Metadata (OMEX)
- COMBINE archive

Specifications of Standards in Systems and Synthetic Biology: Status and Developments in 2022 and the COMBINE meeting 2022. M. König et al. J Integr Bioinform. 2023 Mar 29;20(1).

SBML Level 3: an extensible format for the exchange and reuse of biological models. SM Keating, D Waltemath, M König, ..., M Hucka, and SBML Community members. Mol Syst Biol. 2020;16(8):e9110. doi:10.15252/msb.20199110

The Systems Biology Markup Language (SBML): Language Specification for Level 3 Version 2 Core. M. Hucka, F. Bergmann, C. Chaouiya, A. Dräger, S. Hoops, S. Keating, M. König, N Le Novère, C. Myers, B. Olivier, S. Sahle, J. Schaff, R. Sheriff, L. Smith, D. Waltemath, D. Wilkinson, F. Zhang. J Integr Bioinform. 2019 Jun 20;16(2); doi:10.1515/jib-2019-0021

The Distributions Package for SBML Level 3. L. Smith, S. Moodie, F. Bergmann, C. Gillespie, S. Keating, M. König, C. Myers, M. Swat, D. Wilkinson, M. Hucka. J Integr Bioinform. 2020 Aug 4; doi: 10.1515/jib-2020-0018

Open modeling and exchange (OMEX) metadata specification version 1.0. Neal ML, Gennari JH, Waltemath D, Nickerson DP, König M. J Integr Bioinform. 2020 Jun 25;17(2-3):20200020. doi:10.1515/jib-2020-0020.

Harmonizing semantic annotations for computational models in biology. Neal, M.; König, M.; Nickerson, D., ...mWaltemath, D. Brief Bioinform. 2019 Mar 22;20(2):540-550. doi:10.1093/bib/bby087.



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```
Parameter(  
    "ICGIM_ki_bil",  
    0.02,  
    unit=U.mM,  
    name="Ki bilirubin of icg import",  
    sboTerm=SBO.INHIBITORY_CONSTANT,  
    notes=""  
    bilirubin reference range:  
    ...  
    ~ 0.1 - 5 [g/dL]  
    (10 [mg/L] / 584.6623 [g/mole]) ~ 0.0171 mmole/L  
    ...  
    setting Ki in reference range  
    """,  
    )
```

Parameter	
ICGIM_ki_bil	Ki bilirubin of icg import
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metaID	meta_ICGIM_ki_bil
name	Ki bilirubin of icg import
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value	0.02
constant	✓
units	$\frac{\text{mmol}}{\text{l}}$
derivedUnits	$\frac{\text{mmol}}{\text{l}}$
cvterms	BQB_IS sbo SBO:0000261
inhibitory constant	
Synonym: Ki	
notes	
bilirubin reference range:	
~ 0.1 - 5 [g/dL] (10 [mg/L] / 584.6623 [g/mole]) ~ 0.0171 mmole/L	
setting Ki in reference range	





SBML4Humans: Human readable SBML report

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Parameter

HCT hematocrit

id	HCT
metaID	meta_HCT
name	hematocrit
sbo	SBO:0000002
value	0.51
constant	✓
units	—
derivedUnits	—
cvterms	

BQB_IS sbo SBO:0000002

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Hematocrit

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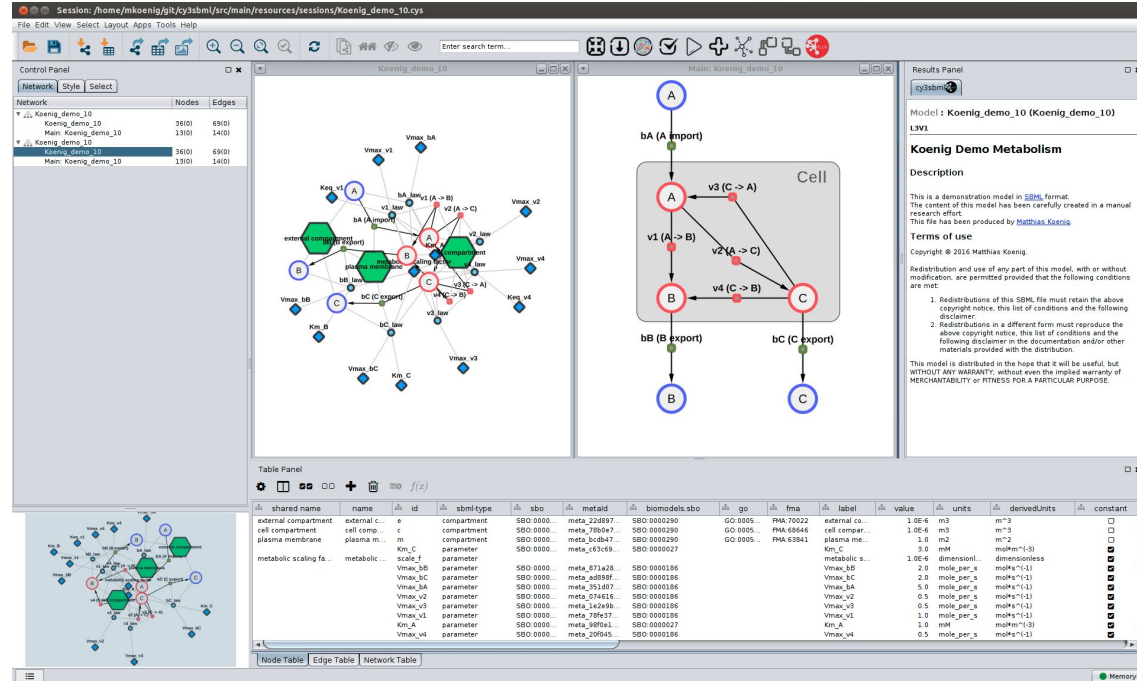
HCT	hematocrit	✓	0.51	—	—	
-----	------------	---	------	---	---	--

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Var	Var		$(1 - HCT) \cdot (BW \cdot FVar - \frac{FVar}{FVar + FVve + FVpo + FVhv} \cdot BW \cdot Fblood \cdot (1 - (FVar + FVve + FVpo + FVhv)))$	<i>t</i>
Vpo	Vpo		$(1 - HCT) \cdot (BW \cdot FVpo - \frac{FVpo}{FVar + FVve + FVpo + FVhv} \cdot BW \cdot Fblood \cdot (1 - (FVar + FVve + FVpo + FVhv)))$	<i>t</i>
Vhv	Vhv		$(1 - HCT) \cdot (BW \cdot FVhv - \frac{FVhv}{FVar + FVve + FVpo + FVhv} \cdot BW \cdot Fblood \cdot (1 - (FVar + FVve + FVpo + FVhv)))$	<i>t</i>
Vre_plasma	Vre_plasma		$Vre \cdot Fblood \cdot (1 - HCT)$	<i>t</i>
Vgi_plasma	Vgi_plasma		$Vgi \cdot Fblood \cdot (1 - HCT)$	<i>t</i>
Vli_plasma	Vli_plasma		$Vli \cdot Fblood \cdot (1 - HCT)$	<i>t</i>
Vlu_plasma	Vlu_plasma		$Vlu \cdot Fblood \cdot (1 - HCT)$	<i>t</i>

cysbml: Cytoscape app for visualizing SBML

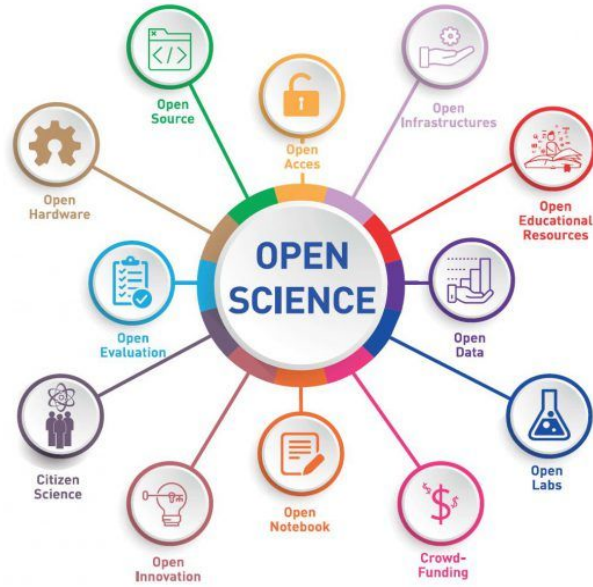
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<https://github.com/matthiascoenig/cy3sbml>



Open und FAIR Science



FE Kohrs, ..., M König, ..., TL Weissgerber. **Eleven Strategies for Making Reproducible Research and Open Science Training the Norm at Research Institutions.** eLife (2023)

Ten Simple Rules for FAIR Sharing of Experimental and Clinical Data with the Modeling Community. König M, Grzegorzewski J, Golebiewski M., Hermjakob H, Hucka M, Olivier B, Keating SM, Nickerson D, Schreiber F, Sheriff R, Waltemath D. Preprints 2021, 2021080303, doi: 10.20944/preprints202108.0303.v2.

FAIR Sharing of Reproducible Models of Epidemic and Pandemic Forecast Ramachandran, K.*; König, M.*; Scharm, M.; Nguyen, T.V.N.; Hermjakob, H.; Waltemath, D.; Malik Sheriff, R.S. (* equal contribution). Preprints 2022, 2022060137 (doi: 10.20944/preprints202206.0137.v1).



Matthias König

Open Data in Repositories

- BioModels DB, GitHub, Zenodo
- PK-DB

Open/FAIR models

- CC-BY, Zenodo, GitHub, FAIR Indicators

Open Science Initiatives

- Strategy development (Open Science Training at Universities)
- Open Science Ambassador (2024-2025)

